ECG Image Digitization

Project Proposal

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Problem Formulation

An electrocardiogram, or ECG, is a non-invasive test used to measure the electrical activity of the heart via electrodes placed on specific locations on the skin [1]. ECGs aid in diagnosing various cardiac conditions such as heart rhythm diseases or even heart attacks as they can provide information on specific heart rates or rhythms [2].

Traditionally, ECGs have been printed on ECG thermal paper. However, the storage of these paper scans does have certain drawbacks, including ink deterioration, folding, overwriting, blurring, and more. Even as hospital charting and data storage largely moves to more digital platforms, ECGs are often uploaded only as images or scans of the paper charts, with the original signal only being preserved on paper records [3]. This is important as images of ECGs are distinct from digital ECG signals. This is because images contain a visual representation of a signal using pixels rather than voltage measurements over time that a digital signal would contain.

Another issue with the sole use of ECG thermal paper or images for storage regards the overall shift to more machine-learning based methods for automatic scan analysis and diagnosis [3]. These techniques require a digital, numerical, version of the exact waveform, all of which cannot readily be obtained from an image scan due to the reason mentioned above. This has the potential to increase wait times and analysis backlogs as practitioners would need to analyze the signals by hand as compared to a model doing the analysis for them.

Even with these drawbacks, ECGs are still printed on ECG thermal paper for several reasons including its familiarity, convenience, and ease of interpretation. In addition, it still serves as a valuable back-up storage option if a hospital were to face any digital outages or malicious attacks. In addition, it is easy for doctors to hand off these scans to other departments as needed, especially during emergencies where time is of the essence. Lastly, it provides a standardized method for ease of interpretation by practitioners. This means that increasing efficiency of any hospital department cannot include the removal of paper ECG scans as a storage option, but it must include a way to accurately convert images of paper ECG scans to a digital waveform that can maintain the signal morphology.

Methods of accurate and efficient conversion of image data to digital waveforms have been attempted through various methods such as traditional image processing utilizing tools like MATLAB to machine learning models, which will be talked about in the prior art section below. However, all these methods have run into similar issues regarding the loss of waveform accuracy and similarity with the original when the input image of the ECG scan was distorted. In addition, these methods also degraded when dealing with real-world photographs rather than an image scan due to the increased chance of distortions being present. In addition, as with all of these prior methods, it is important to determine metrics to quantify similarity between the reconstructive waveform and the original waveform. Previous methods used metrics such as Pearson's correlation coefficient and signal to noise ratio to do so.

Our team seeks to replicate and improve the above systems, with a focus on maintaining reconstruction accuracy with poor quality inputs, such as real-world photos, or distorted images.

Overall, the project has three aims:

1. To establish a reliable process for converting images of paper ECG scans to digital numerical waveforms while preserving the morphology of the original signal

2. To evaluate the dependability of this process when images of ECG scans are not ideal, meaning that the images are distorted by any of a number of factors including, handwritten notes, rotations, shadows, wrinkles, blurs, and other distortions.

3. To use the failure modes found in Aim 2 to improve the strongest published model, Signal-Savants, as measured by SNR and Pearson correlation.

Prior Art

Beginning in 2005 with the development of the initial version of the patented ECGscan system [4] through 2021 with the publication of an open source alternative [5], researchers have devised various semi-automatic methods for converting ECGs stored on paper to digital time series. Both ECGscan and the open-source method are able to account for some photographic distortions such as cropping and rotation, yet could not account for damage to the paper ECGs.

The open-source semi-automatic method requires all images to be scanned in the same way, then for rotation, lead locations, lead start time, and grid scale to be input manually. Once the images are preprocessed this way, they are converted to a binary image with each pixel assigned one of two possible values. The pixels belonging to the grid lines and waveform are extracted. The grid lines are then used to scale the voltage and time dimensions of the wave form to derive the final time series.

Rapid development of machine learning technology since 2021 has led to both increased use cases for large numbers of digitized ECGs as well as the potential to fully automate ECG digitization [6]. Training a model to automatically digitize ECG requires both the target signals as time series and distorted images of paper ECG for each target signal. Similarly to semi-automatic ECG digitization methods [4] [5], it would be possible to train and test a model on images of physical paper ECGs printed from a dataset of ECG time series. However, those images lack realistic varied distortions found in images of existing paper ECGs because of varying photography and storage conditions. As such, existing automatic ECG image digitization methods use computationally generated replicas of images of paper ECGs [7].

ECG Image Kit is an open source toolkit used to generate training images from ECG time series. Images generated this way are used to train most state-of-the-art digitization methods. It first generates an undistorted image from ECG time series data. Beginning by picking a random size and resolution, vertical and horizontal gridlines are added at either a 1mm or 5mm interval with each mm vertically representing 0.1mV and each mm horizontally representing 40ms. The ECG waveforms can then be plotted scaled to those dimensions. Finally header and label information is displayed. Header text including information like date, sex, and age may be displayed as is available with the ECG data. Each lead in the ECG may be labeled with an index number and points of interest in time may be marked [7].

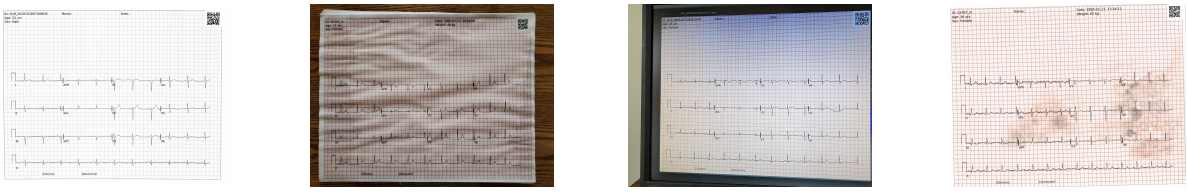
The following distortions will then be applied to the image to a variable extent: handwritten notes, wrinkles, creases, rotation, crop, noise, color temperature. Wrinkles are pseudo-random folds in the paper and are generated from a set of 20 transparent grayscale images of papers with different wrinkle patterns which are combined with image quilting to generate unique patterns. Creases are straight lines crossing the image which have been blurred to replicate the shadow left by creasing paper [7].

The George B. Moody PhysioNet Challenge in 2024 tackled ECG digitization, and participants developed methods to convert images of paper ECGs to digital signal data. Sixteen groups participated and submitted their model to be scored. The training set was built from the PTB-XL dataset, containing 21,799 12-lead ECG recordings; teams could optionally add the PhysioNet Challenge 2021 datasets (88,253 recordings) and the CODE-15% dataset (345,779 recordings) by converting them to images with ECG Image Kit [8]. The submissions were evaluated on their performance digitizing a hidden dataset of images to the hidden set of time series associated with those images. However, while the models would be trained on only generated images, the test set would contain some number of real images as well. The performance of each submission was evaluated using signal to noise ratio (which we will also be using, explained in depth in the Evaluation Plan section below).

The winner of this challenge, having been scored with a signal to noise ratio of 12.15, was team SignalSavants from Oxford [8]. They trained their model on images generated using ECG Image Kit from the PTB-XL dataset. This group's model begins by identifying the gridlines and using them to rotate the images until the horizontal gridlines are horizontal and the vertical gridlines

are vertical. They trained nnU-Net on their dataset to generate a pixel mask over each of the waveforms. For each column of pixels in the mask, the average height of the pixels in that column is added to a time series. Using the grid lines, the number of values in that time series, and magnitude of each value are scaled to match the signal. SignalSavants independently evaluated the signal to noise ratio of their model and found it to be 17.02 [9].

Two other methods are also replicated in this project: the classical image-processing pipeline of Baydoun et al. [10], and WAVIE [11], a modular open-source Python pipeline from the same Challenge that combines learned orientation correction, nnU-Net segmentation, YOLOv5-based lead localization, and OpenCV signal extraction. Both are described further in the Proposed Method section.



(a) Clean (b) Wrinkled (c) Photographed (d) Water-damaged

Figure 1: The same record from the ECG-Image-Database [12] in four variants, from a clean printout to real physical damage.

The images generated with ECG Image Kit do not capture the full extent of distortions in the real world. There are many variations of image distortion which may exist in images of real paper ECGs, but which ECG Image Kit is not capable of replicating or not capable of replicating well. Some of these issues have been resolved with modified versions of the toolkit, such as in Stenhede et al [6], where perspective distortion, such as from an image taken with a phone camera is simulated. Additionally we are concerned about reading paper ECGs with water damage, faded ink, redacted information, image compression artifacts, as the toolkit has no solution for generating these. The ECG-Image-Database [12] covers many of these artifacts with physically produced damage (Figure 1), which is why we evaluate on it directly. And in places where the database and the kit leave gaps (such as creases that bend the waveform in three dimensions), we hope to extend the capabilities of ECG Image Kit as a reach goal (see Expected Results).

Proposed Method

This project aims to establish a reproducible workflow for ECG image digitization, converting scanned or photographed paper-based 12-lead electrocardiogram (ECG) images into digital ECG time-series with accurate time axes and voltage units. The research focus is not on developing entirely new deep learning architectures but rather on using existing classical image-processing methods as a baseline to further compare and improve the reconstruction accuracy and robustness of learning-based approaches under varying image conditions. The overall process is divided into four steps: paired data preparation, waveform reconstruction, signal calibration, and quantitative evaluation. First, ECG images and their corresponding original digital ECGs are obtained; then, waveforms are extracted through image processing or deep learning; next, pixel coordinates are converted into time and voltage; finally, the reconstruction quality is evaluated using ground-truth waveforms. This research design aligns with the ECG digitization methods proposed by Baydoun et al. (2019) and in the 2024 PhysioNet Challenge [9–11].

Reliable, paired image-signal data serves as the foundation for model development and evaluation. Real paper-based ECGs typically retain only the images but lack the original digital signals, making it difficult to directly calculate waveform reconstruction error. To address this issue, we use the ECG-Image-Database [12], in which each image is paired with its original digital ECG, as our

primary data source (see Data). Its synthetic images were rendered from authentic 12-lead ECG recordings, such as PTB-XL, using ECG-Image-Kit, with standard ECG grids, lead labels, and paper-style layouts [7, 13]. The underlying signals originate from digitally measured ECGs, with only the image presentation form and added visual distortions generated by software. Therefore, each synthetic image can retain the corresponding original digital ECG as ground truth, providing reliable references for supervised model development and quantitative evaluation. However, synthetic images cannot fully simulate real paper damage, camera perspective, uneven illumination, and scanning artifacts. To evaluate synthetic-to-real generalization, we also use the database's real scanned, photographed, and physically distorted images. In addition, we plan to use the ECG-Image-Kit to generate varying levels of noise, rotation, wrinkles, creases, and other image artifacts from PTB-XL records to establish controlled testing conditions. First, we will validate the baseline algorithm using clean images, then gradually introduce image distortions to observe how the reconstruction performance of different methods changes.

This project first plans to reproduce the classical ECG digitization algorithm proposed by Baydoun et al. (2019) and use it as the classical baseline for subsequent comparisons [10]. The method primarily consists of five processing steps. First, detect the Region of Interest (ROI) for each ECG lead using local pixel-intensity variation to segment the complete image into signal regions suitable for independent processing. Second, perform image binarization via Otsu Thresholding to separate foreground pixels potentially belonging to ECG waveforms from the background. Third, group interconnected foreground pixels using Connected Component Analysis to filter out isolated noise and irrelevant artifacts. Fourth, the Hough Transform is utilized to assist in determining the waveform reference centerline, and combined with connected-pixel tracking, the ECG trace is restored along the image columns. Finally, missing waveform points are filled through interpolation, and moderate smoothing is applied to reduce extraction noise while preserving key morphological features such as the QRS complex and R peaks as much as possible. After completing waveform extraction, we will convert pixel coordinates into digital signals represented in seconds and millivolts based on the ECG grid scale and available calibration metadata. During implementation, we will inspect ROI bounding boxes, binary masks, and reconstructed waveform overlays to identify errors such as grid interference, broken traces, and peak distortion.

After establishing the classical baseline, we plan to further investigate two learning based methods in the 2024 George B. Moody PhysioNet Challenge: SignalSavants and WAVIE. SignalSavants was proposed by Krones et al. (2024), which combines Hough based rotation correction, nnU Net segmentation, and mask vectorization to identify waveform pixels and recover numerical signals from ECG images [9]. WAVIE was proposed by Verlyck et al. (2024) and adopts a modular architecture, including learned orientation correction, nnU-Net waveform segmentation, YOLOv5-based lead localization, and signal extraction based on OpenCV [11]. Our implementation strategy is to prioritize evaluating the public implementations and available pre-trained weights of these methods, confirming their compatibility with the target data, and then deciding whether they need to be retrained or adapted. This can reduce unnecessary training costs and focus research on reconstruction reliability and method comparison. Due to the different intermediate outputs generated by the three methods, such as pixel paths, segmentation masks, or lead bounding boxes, we will establish a unified output representation. Each method should ultimately produce numerical ECG waveforms with correct lead identity, time axis, and voltage calibration. All methods will use consistent signal alignment, missing sample handling, and evaluation protocol on the same held out ECG records to ensure fairness in comparison.

Once this comparison shows the failure modes for every baseline, we plan to make edits to the SignalSavants model, the strongest of the three, to improve SNR and/or Pearson correlation on those failure modes. Candidates include changes to the training data, the training recipe (augmentation, the mix of clean and damaged images), and to individual pipeline stages (rotation correction, segmentation, vectorization). Each edit will be compared against an unmodified SignalSavants model retrained on our training split with the same budget, one at a time, to test any gains clearly.

Data

Dataset. We use the ECG-Image-Database [12], which pairs images of 12-lead ECG printouts with the reference digital signals they were made from (WFDB format). It has 37,191 images (about 37 GB) from 2,243 records: PTB-XL (977) [13], Emory Healthcare (1,000), and Akershus University Hospital (266). The input to every method is one image and the output is the 12-lead time series, which is scored against the reference signal. The database is a later, larger release than the Challenge data (the 2024 Challenge trained on PTB-XL and tested on hidden real scans from additional databases), and subsets of it were used in the 2024 and 2025–2026 Challenges, so our scores are not directly comparable to published Challenge scores.

Variability. Each PTB-XL and Emory record appears in about 18 of the database's 28 image variants (35,595 images from 1,977 records). These are electronic printouts rendered with ECG-Image-Kit [7], programmatic variants such as rotation, creases, wrinkles, and simulated handwriting, and physically printed copies that were scanned, photographed (including photographs of computer monitors), or deliberately damaged (water, coffee, sauce, heat, humid storage, mold). The Akershus images are different in kind: real clinical thermal-paper ECGs [6] captured with a flatbed scanner and two phones (1,596 images), with a different layout from the other two sources: the full 5 s of all 12 leads is shown across two separate images that must be associated correctly. We group the images as synthetic (electronic and programmatic) and real (physically printed, photographed, damaged, and Akershus), and report Akershus separately.

Known limitations. Most PTB-XL and Emory leads show only 2.5 s of the original 10 s on the printout, so the remainder cannot be recovered from the image. SignalSavants' public weights were trained on PTB-XL, so its PTB-XL scores may be optimistic; we report PTB-XL separately from Emory and Akershus. PTB-XL and Emory images share one layout and one physical-damage procedure. Akershus comes from a single hospital with only 266 records. The diagnostic labels we know of are the PTB-XL superclasses (normal, myocardial infarction, ST/T change, conduction disturbance, hypertrophy), so diagnostic breakdowns are limited to PTB-XL unless labels for the other sources can be obtained.

Development and evaluation. All splits are made by record (70/15/15 train/validation/test), so no variant of a record appears on both sides. We develop on a 20% record-level subset and make all design decisions on validation records. Test records are used once, for the final results. Akershus is held out entirely as an unseen source and is never used for training or tuning. For the graded-severity analysis we also generate images from PTB-XL records with the existing ECG-Image-Kit, which adds increasing noise, rotation, wrinkles, and creases to otherwise clean renderings.

Evaluation Plan

Our primary metric is the official PhysioNet Challenge 2024 signal-to-noise ratio (SNR), the same metric SignalSavants was scored with. In this context the signal is the original ECG time series and the noise is the difference between the signal and the estimated time series after applying an adjustment for a minor offset. Let y_i be the signal at time step i and let $\hat{y}_i$ be the estimated time series at time step i. The equation for signal to noise ratio is the following [8]:

$$\text{SNR (dB)} \;=\; 10\,\log_{10}\left(\frac{\sum_i y_i^2}{\sum_i (y_i - \widehat{y}_i)^2}\right)$$

The digitized signal is first aligned to the reference (up to ±0.5 s and ±1 mV), the score is scaled by the fraction of samples digitized, and scores are averaged across records and leads. Trained models will be reported as a mean over 3 seeds plus bootstrap confidence intervals over records, since three seeds alone say little about variance.

We will also evaluate the Pearson Correlation Coefficient for every method in comparison to the true waveforms, computed after the same alignment as SNR. This is a standard metric for comparing

waveforms and was used by Baydoun et al. [10] in their method. Pearson correlation is calculated as follows:

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2}\sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}$$

Baydoun et al., SignalSavants, and WAVIE will be compared on the same held-out records (minimum deliverable), and the edited model will then be compared against the reproduced SignalSavants baseline (target). The reproduced baseline will be scored with the same splits, scoring routine, and seeds as every edited model, and edits are tested one at a time (see Proposed Method). Splits and held-out data are described in the Data section.

The evaluation will be conducted in order of difficulty: first, use clean synthetic images to verify the correctness of the algorithm; next, test robustness under controlled distortions; finally, evaluate real-world generalization using physically scanned or photographed ECG images. We also plan to conduct subgroup analysis based on distortion type and ECG diagnostic category, where data and labels allow, to investigate whether image quality and waveform morphology affect reconstruction accuracy. Through unified data partitioning, signal calibration, and evaluation procedures, this project aims to establish a reproducible ECG digitization comparison framework that systematically identifies the advantages, limitations, and main failure modes of classical and learning based methods under different conditions.

Expected Results and Deliverables

We plan to organize the project into five stages over the year: replicate the published methods, measure where it fails, edit the model to address those failures, and report final results on both synthetic and real images. Stages 1 and 2 are our minimum deliverable, resulting in a reproducible comparison of Baydoun et al., SignalSavants, and WAVIE, with a subgroup analysis showing where each method's strengths and weaknesses lie. Stages 3 and 4 are our target, where we improve the model and measure the change in SNR and Pearson correlation. Extending ECG Image Kit is a reach goal.

Table 1: Project stages, primary metrics, and success targets.

Stage	**Timeline**	**Primary metric**	**Target**
1. Baseline Replication (minimum)	Fall weeks 8-11	Official SNR	Replicate the three methods and land near their published results.
2. Subgroup Analysis (minimum)	Fall weeks 11-13	SNR and Pearson by distortion type and diagnostic class	Rank failure modes with confidence intervals
3. Model Improvement (target)	Fall week 13 to spring week 5	SNR and Pearson on validation records, by failure mode	Produce improvement beyond noise on targeted failures
4. Final Results (target)	Spring weeks 6-8	SNR and Pearson on held-out test records, synthetic and real	Gain beyond noise in SNR and/or Pearson on the targeted failures, on both synthetic and real images, or a documented split
5. Dissemination (target)	Spring weeks 9-14	Reproducibility of reported results	Final report, documented codebase, 30-minute talk, and Design Day poster
Reach Goal: Extend ECG-Image-Kit	Spring weeks 1-8	SNR and Pearson by distortion type and severity	New distortions expose failures that existing images do not, or a documented null result

Stage 1: Baseline Replication (Fall weeks 8-11)

Deliverable. Working versions of all the tested methods that output waveforms in one shared format (with lead identity, time axis, and voltage scale), plus the data loader and record-level splits. SignalSavants and WAVIE are run from their public code and pre-trained weights and retrained only if the weights don't work for us. Baydoun et al. is implemented from the paper.

Success criterion. Each method's score on our evaluation should fall near its published results (for SignalSavants, in the range bracketed by its two reported scores, 12.15 on the Challenge test set and 17.02 in their own evaluation), or we can explain why it does not. Every later claim of improvement is measured against the reproduced SignalSavants baseline.

Fallback. If we cannot reach that range, we will retrain, report the gap, run an ablation to locate its source, and adopt our best reproduction as the baseline. A method that cannot be replicated in time is dropped from the comparison, with the reason documented, but we aim to keep at minimum SignalSavants plus one other method.

Stage 2: Subgroup Analysis (Fall weeks 11-13)

Deliverable. SNR and Pearson correlation for all three methods as tested on the ECG-Image-Database, broken down by distortion type and by PTB-XL diagnostic class. We also generate graded-severity images from PTB-XL with ECG-Image-Kit, adding noise, rotation, wrinkles, and creases in steps, to see how performance drops as the distortion increases.

Success criterion. A ranked list of failure modes with confidence intervals, which tells us what Stage 3 changes should aim to fix.

Fallback. If diagnostic labels cannot be obtained for Emory and Akershus, the diagnostic breakdown is restricted to PTB-XL and the distortion breakdown covers all three sources.

Stage 3: Model Improvement (Fall week 13 to spring week 5)

Deliverable. Our edited version of the SignalSavants model, with edits chosen from the failure modes found in Stage 2. The candidate edits and how they are tested are described in the Proposed Method section.

Success criterion. We cannot promise a specific dB gain, since Stage 2 will show how much headroom we even have. But success means an improvement in SNR and/or Pearson correlation on the targeted failure modes, on validation records, larger than the variation across seeds and bootstrap resamples.

Fallback. If no edit helps, we will create a record of which edits were tried and why they failed.

Stage 4: Final Results (Spring weeks 6-8)

Deliverable. A final document comparing the reproduced SignalSavants baseline and our edited model on synthetic images and on real images, with per-distortion and per-diagnostic-class breakdowns and confidence intervals, computed once on held-out test records.

Success criterion. Our model improves on the reproduced SignalSavants baseline beyond noise in SNR and/or Pearson correlation on the targeted failure modes.

Fallback. If gains appear on only one of the failure modes or data types, we report that split. A gain on synthetic but not real images would document the synthetic-to-real gap.

Stage 5: Dissemination (Spring weeks 9-14)

Deliverable. The final report covering both semesters, a 30-minute oral presentation, a poster for BME Design Day, and a finalized code base with well-commented functions, documentation of

each function, scripts that reproduce every table and figure, an environment file, the exact splits and seeds, and a README.

Success criterion. A team member who did not write a given component can reproduce the headline numbers from a clean clone using the README alone.

Reach Goal: Extend ECG-Image-Kit (Spring weeks 1-8, started only once Stage 3 is on schedule). The deliverable is a modified version of ECG Image Kit that adds distortions the kit cannot currently produce and the database does not cover, starting with creases and wrinkles that bend the waveform, gridlines, and text in three dimensions. If time does not allow, the reach goal is dropped without affecting Stages 1-5 and documented as future work.

Expected Pitfalls and Mitigation Strategies

Compute limitations (Stages 1-4). We have access to a cluster as confirmed by the instructors, but a shared cluster might mean queue delays, storage and GPU-hour limits, and preemption. We will run all candidate edits on a development subset of the data and reserve full-scale training for the baseline and the final edited model. We will checkpoint every run to ensure no need for unnecessary retraining.

Replicating existing model accuracy (Stage 1). SignalSavants' score depends heavily on the test set: 12.15 on the Challenge's hidden set, which included real images, versus 17.02 in their own evaluation. Our number on the ECG-Image-Database will not match either exactly. Therefore, success will be to fall in that range or around it. To ensure we do not spend too long chasing the exact numbers, we fix splits and seeds from day one, use the official SNR implementation, and time-box replication to the end of fall week 11, then proceed with our best version and document the gap.

Improving on an already accurate model (Stage 3). The reproduced SignalSavants baseline is already stronger than competitors by a large margin and was built with only about 20k training images and simple masks. We are not sure what the headroom for improvement will be. Therefore, we will keep edits small and targeted, test them one at a time at a fixed budget, so that any difference is attributable to the edit and not to extra capacity or data.

Small subgroups (Stages 2-4). Akershus has only 266 records from one hospital, and the diagnostic labels we know of come from PTB-XL only. Splitting by both distortion type and diagnosis means it's likely that some of the subgroups have very few samples. We will use bootstrap confidence intervals and merge sparse cells.

Broader Impacts

As the medical field increasingly moves to digital platforms for both storage and analysis, it is imperative that ECG data goes along with it. Digitizing ECG signals has two major benefits in the broader medical field, reconstruction of older ECG data for retrospective research, and the support of automated signal analysis and machine-learning based diagnosis. Digitization can lead to an increase in available data for machine learning models and for researchers to use for more targeted studies about cardiovascular conditions. However, this entirely depends on the quality of the reconstructed signal and the potential accuracy loss when coming into contact with distorted image inputs. Therefore, this work and further evaluation metrics of digitization models remain important to ensure that reconstructed signals are accurate and trustworthy for downstream processing and analysis.

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Contributions

Bohan: Researched and wrote the Proposed Method, Data, and Evaluation Plan sections
Ansh: Wrote the Expected Results and Expected Pitfalls sections, edited the final document
Mateo: Researched and wrote the Prior Art and Evaluation Plan sections
Suruthikha: Researched and wrote the Problem Formulation and the Broader Impact sections
AI use disclosure: Claude was used to help clean up the writing and fix wording mistakes or inconsistencies. It was also used to translate the document from Word to LaTeX.